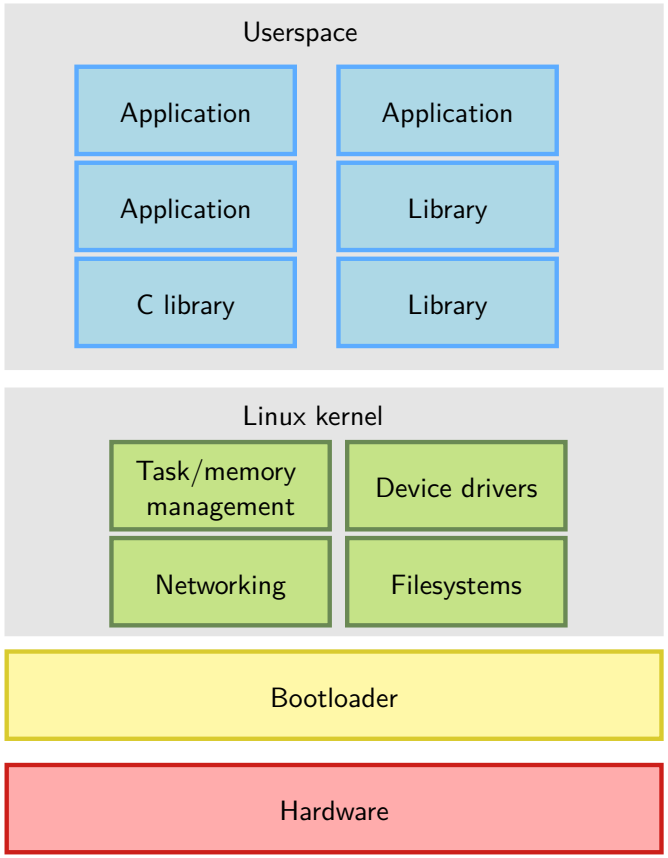
Introduction to Embedded Linux



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Corrections, suggestions, contributions and translations are welcome!

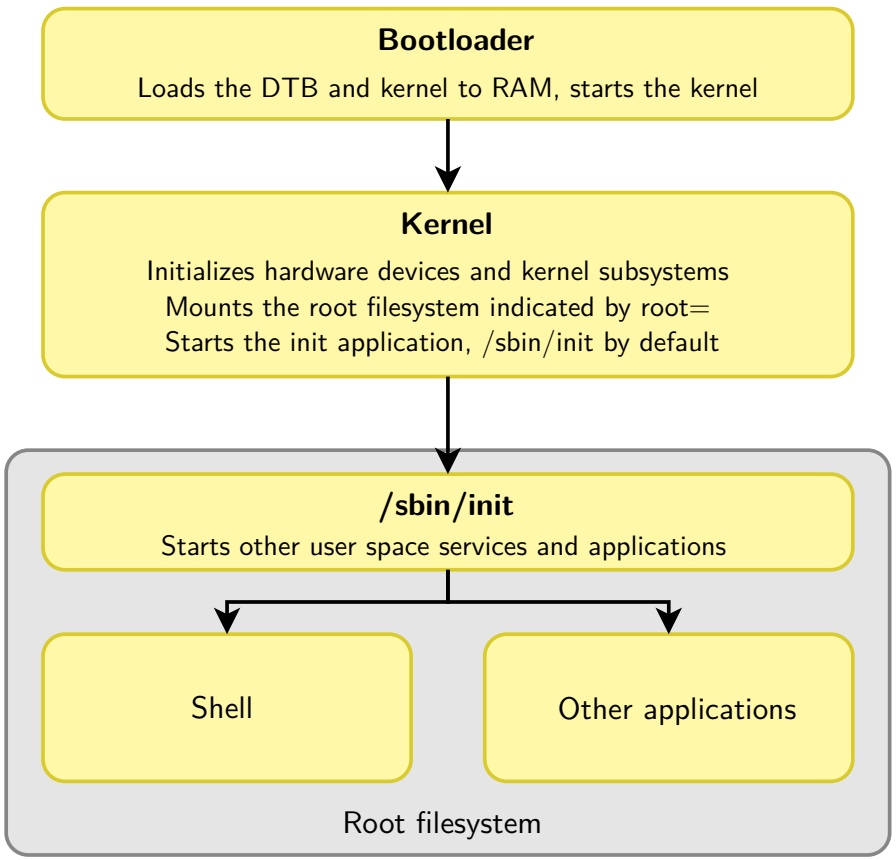


Simplified Linux system architecture





Overall Linux boot sequence

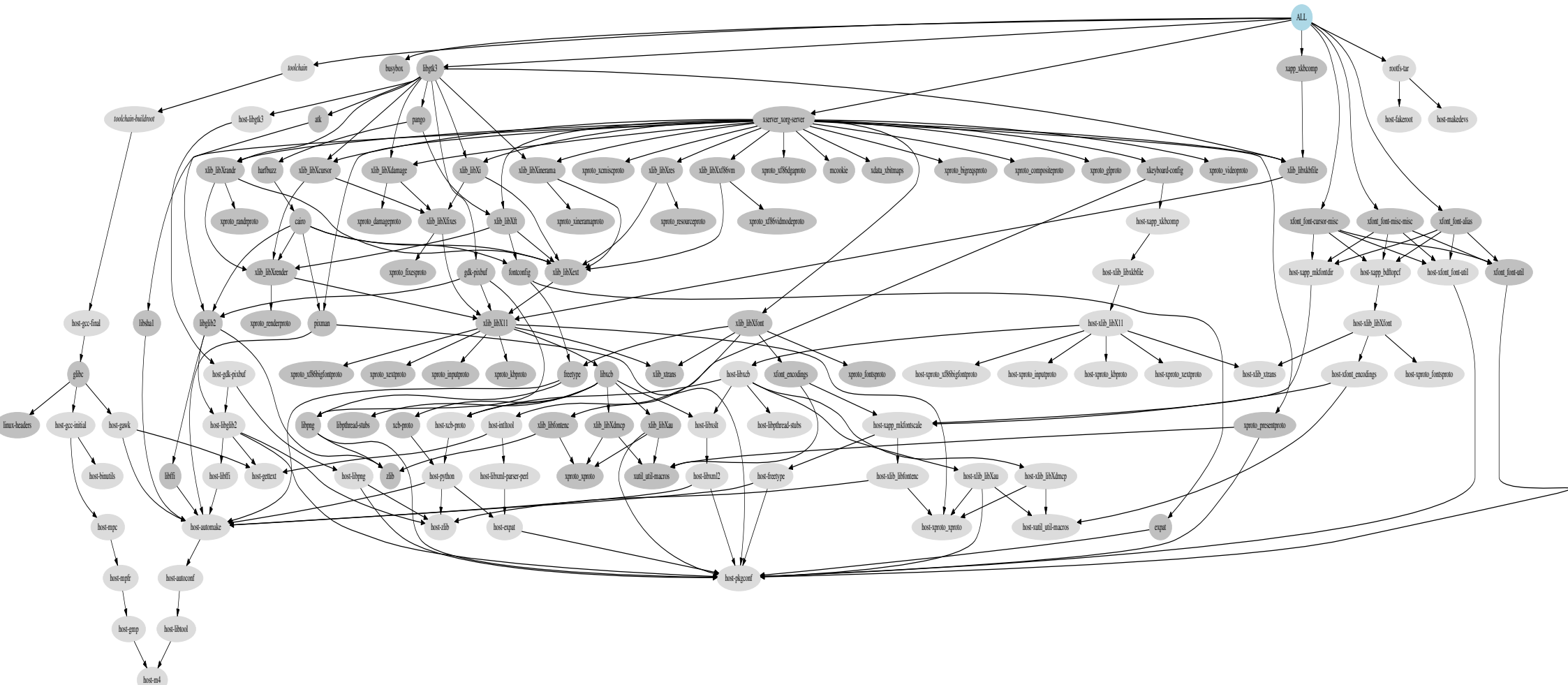




Embedded Linux work

- **BSP work**: porting the bootloader and Linux kernel, developing Linux device drivers.
- **system integration work**: assembling all the user space components needed for the system, configure them, develop the upgrade and recovery mechanisms, etc.
- **application development**: write the company-specific applications and libraries.

Complexity of user space integration

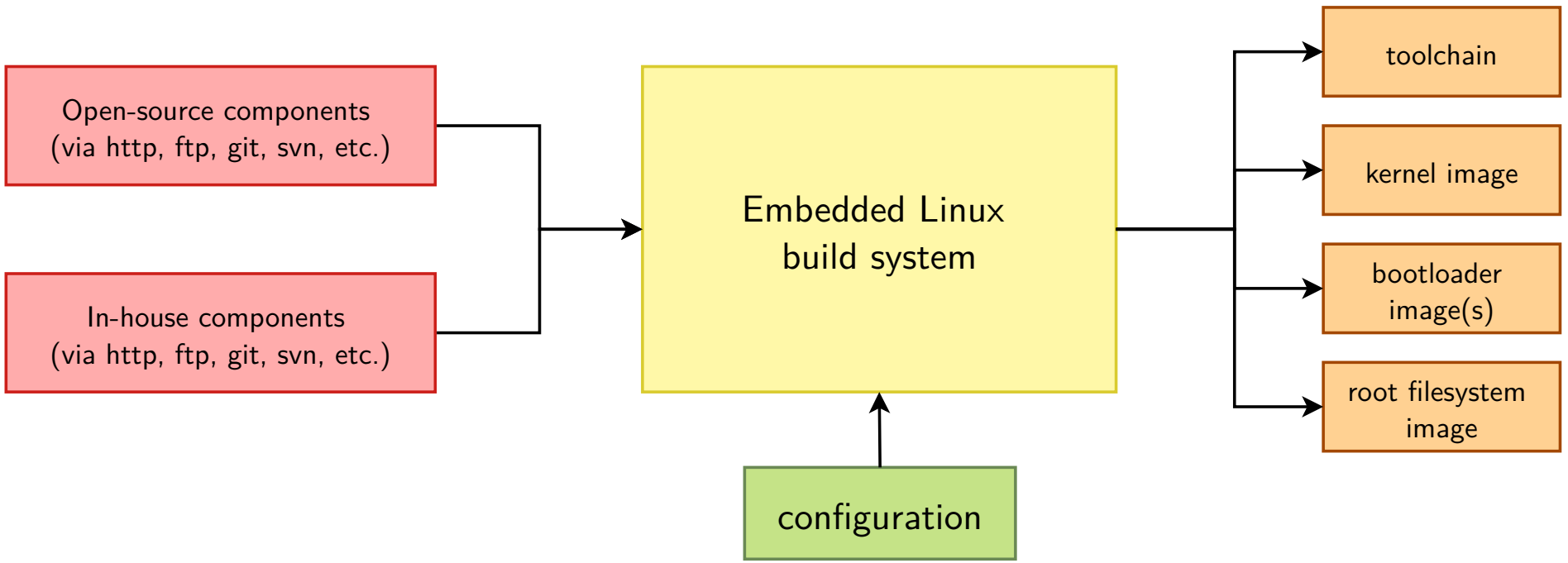




System integration: several possibilities

	Pros	**Cons**
Building everything manually	Full flexibility Learning experience	Dependency hell Need to understand a lot of details Version compatibility Lack of reproducibility
Binary distribution Debian, Ubuntu, Fedora, etc.	Easy to create and extend	Hard to customize Hard to optimize (boot time, size) Hard to rebuild the full system from source Large system Uses native compilation (slow) No well-defined mechanism to generate an image Lots of mandatory dependencies Not available for all architectures
Build systems Buildroot, Yocto, PTXdist, etc.	Nearly full flexibility Built from source: customization and optimization are easy Fully reproducible Uses cross-compilation Have embedded specific packages not necessarily in desktop distros Make more features optional	Not as easy as a binary distribution Build time

Embedded Linux build system: principle



- Building from source → lot of flexibility
- Cross-compilation → leveraging fast build machines
- Recipes for building components → easy



Embedded Linux build system: tools

- A wide range of solutions: Yocto/OpenEmbedded, PTXdist, Buildroot, OpenWRT, and more.
- Today, two solutions are emerging as the most popular ones
  - **Yocto/OpenEmbedded** Builds a complete Linux distribution with binary packages. Powerful, but somewhat complex, and quite steep learning curve.
  - **Buildroot** Builds a root filesystem image, no binary packages. Much simpler to use, understand and modify.